

# 字符串练习3

## 人员

臧晨云 隋湫琛 于昊华 初星熠 刘奕男 张高 陈之恒 霍佳铭 李景翊 刘子彬 到课

## 上周作业检查

上周作业链接: <https://cppoj.kids123code.com/contest/4521>

### 王向东老师周六一点半C++字符串练习2

🔄 刷新

| #  | 用户名          | 姓名  | 编程分 | 时间 | A   | B   | C   | D   | E   | F   |
|----|--------------|-----|-----|----|-----|-----|-----|-----|-----|-----|
| 1  | zangchenyun  | 臧晨云 | 600 | 70 | 100 | 100 | 100 | 100 | 100 | 100 |
| 2  | zhanggao     | 张高  | 600 | 75 | 100 | 100 | 100 | 100 | 100 | 100 |
| 3  | hepeiyu      | 何佩瑜 | 600 | 76 | 100 | 100 | 100 | 100 | 100 | 100 |
| 4  | chenzhiheng  | 陈之恒 | 600 | 76 | 100 | 100 | 100 | 100 | 100 | 100 |
| 5  | huojiaming   | 霍佳铭 | 600 | 77 | 100 | 100 | 100 | 100 | 100 | 100 |
| 6  | liuzichen123 | 刘子彬 | 600 | 77 | 100 | 100 | 100 | 100 | 100 | 100 |
| 7  | suihaochen   | 隋湫琛 | 600 | 79 | 100 | 100 | 100 | 100 | 100 | 100 |
| 8  | yuhaochua    | 于昊华 | 500 | 75 | 100 | 100 | 100 | 100 | 100 | 0   |
| 9  | chuxingyi    | 初星熠 | 500 | 76 | 100 | 100 | 100 | 100 | 100 | 0   |
| 10 | liuyinan     | 刘奕男 | 500 | 76 | 100 | 100 | 100 | 100 | 100 | 0   |
| 11 | liukeyang    | 刘克阳 | 500 | 76 | 100 | 100 | 100 | 100 | 100 |     |
| 12 | lijingxuan   | 李景翊 | 500 | 77 | 100 | 100 | 100 | 100 | 100 |     |

## 作业

<https://cppoj.kids123code.com/contest/4673> (课上讲了 A ~ F 题, 作业是 A ~ F 题必做, G 题选做)

## 课堂表现

今天重点给同学们讲解了字符串的 substr 和 find 两个方法, 第四题是一道非常综合的练习题, 建议同学们课下多写几遍第四题。

## 课堂内容

### 忽略大小写的字符串比较

```
#include <bits/stdc++.h>

using namespace std;

int main()
{
    string a, b;
    getline(cin, a);
    getline(cin, b);
    int n = a.size();
```

```
for (int i = 0; i < n; i++) {
    if (a[i]>='a' && a[i]<='z') {
        a[i] = a[i] - 'a' + 'A';
    }
    if (b[i]>='a' && b[i]<='z') {
        b[i] = b[i] - 'a' + 'A';
    }
}

if (a < b) cout << "<";
else if (a > b) cout << ">";
else cout << "=";
return 0;
}
```

## 调换位置

```
#include <bits/stdc++.h>

using namespace std;

int main()
{
    string s;
    cin >> s;
    int n = s.size(), p;
    for (int i = 0; i < n; ++i) {
        if (s[i] == ',') {
            p = i;
            break;
        }
    }

    cout << s.substr(p+1) << "," << s.substr(0, p);
    return 0;
}
```

## 验证子串

```
#include <bits/stdc++.h>

using namespace std;

int main()
{
    string a, b;
    cin >> a >> b;
    if (a.find(b) != -1) {
```

```
    cout << b << " is substring of " << a << endl;
}
else if (b.find(a) != -1) {
    cout << a << " is substring of " << b << endl;
}
else {
    cout << "No substring" << endl;
}
return 0;
}
```

### 【深基6.例6】文字处理软件

```
#include <bits/stdc++.h>

using namespace std;

int main()
{
    int n;
    cin >> n;
    string s;
    cin >> s;
    for (int i = 1; i <= n; ++i) {
        int op;
        cin >> op;
        if (op == 1) {
            string t;
            cin >> t;
            s += t;
            cout << s << endl;
        }
        else if (op == 2) {
            int a, b;
            cin >> a >> b;
            s = s.substr(a,b);
            cout << s << endl;
        }
        else if (op == 3) {
            int a;
            string t;
            cin >> a >> t;
            s = s.substr(0,a) + t + s.substr(a);
            cout << s << endl;
        }
        else {
            string t;
            cin >> t;
            int p = s.find(t);
            cout << p << endl;
        }
    }
}
```

```
    }  
    return 0;  
}
```

## atoz

```
#include <bits/stdc++.h>  
  
using namespace std;  
  
int main()  
{  
    string s;  
    cin >> s;  
    int n = s.size();  
    s = " " + s;  
  
    int l, r;  
    for (int i = 1; i <= n; ++i) {  
        if (s[i] == 'A') {  
            l = i;  
            break;  
        }  
    }  
    for (int i = n; i >= 1; i--) {  
        if (s[i] == 'Z') {  
            r = i;  
            break;  
        }  
    }  
  
    cout << r-l+1 << endl;  
    return 0;  
}
```

## 删除单词后缀

```
#include <bits/stdc++.h>  
  
using namespace std;  
  
int main()  
{  
    string s;  
    cin >> s;  
    int n = s.size();  
    if (n>=2 && s.substr(n-2)=="er") {  
        cout << s.substr(0, n-2) << endl;  
    }
```

```
}  
else if (n>=2 && s.substr(n-2)=="ly") {  
    cout << s.substr(0, n-2) << endl;  
}  
else if (n>=3 && s.substr(n-3)=="ing") {  
    cout << s.substr(0, n-3) << endl;  
}  
else {  
    cout << s << endl;  
}  
return 0;  
}
```